

FACS coding of televised darts players reveals different facial behavior before successful and unsuccessful performances

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ABSTRACT

The main objective of the present research was to use FACS on the televised pre-performance facial behavior of professional darts players to investigate if the facial behavior differed between successful and unsuccessful performances. A secondary goal of this study was to test if manual facial coding differed from automated coding using Noldus FaceReader. Therefore, a random sampling methodology was utilized to assemble video clips of the facial behavior of all players immediately before throwing darts in the 2017 World darts Championship in two performance categories, poor and good performance (10 clips per players respectively). Two FACS-certified coders blind to performance condition coded the activation intensity of 20 distinct muscle groups in the face (i.e., action units) and were compared to Noldus FaceReader output. Results showed very limited facial muscle movements in the videos across both performance categories and hardly any correlation between FACS and FaceReader data. However, multivariate data analysis of the FACS data revealed less intense action unit activity before good performance vs. poor performance. More specifically, successful darts performance was associated with lower muscle activity around the eyes and the mouth. These findings are discussed in light of previous literature suggesting that the face can be indicative of levels of arousal and anxiety and that low levels of arousal and anxiety prior to fine sensorimotor coordination are associated with performance in these situations.

The human face is believed to convey an incredible amount of information in social interactions. An important methodological milestone in measuring facial behavior was the development of the Facial Action Coding System (FACS; Ekman & Friesen, 1976, 1978; Ekman et al., 1971) that subsequently led to a rapid accumulation of knowledge regarding the importance of the face in communicating information about internal states such as emotions, attitudes or social intentions (Matsumoto & Hwang, 2013). While the FACS has mainly been used in the study of emotion, it has also been used to detect other internal states like pain or concentration as muscle movement in the face can be indicative of other internal states besides emotions (e.g., Rozin & Cohen, 2003). The main objective of the present research was to use FACS on the televised pre-performance facial behavior of professional darts players to investigate if the facial behavior differed between successful and unsuccessful performances. A secondary goal of this study was to test if manual facial coding differed from automated coding using Noldus FaceReader.

Pertinent to the present research, it has been argued that people utilize facial expressions (amongst further bodily cues) to predict how other people are likely to behave (e.g., Crivelli & Fridlund, 2018; Hall

et al., 2019). Important terms in the field of nonverbal communication that are relevant to the present research are signals versus signs or cues that are given versus cues that are given off (Goffman, 1959). The term signal typically is used to refer to behaviors that belong to a consensually understood code that are shown intentionally (e.g., winking one eye when telling a joke), whereas signs or cues refer to behavior that might be informative but unintended. Although it is problematic establishing intentionality and determining if sender and receiver of nonverbal information share a common understanding of this information, the present research utilizes these common terms in the literature to clarify the focus of this study on unintentional nonverbal cues that might be indicative of successful and unsuccessful sensorimotor performance in darts.

Prior research has shown that pre-performance NVB of professional darts players gives valid information to observers about subsequent performance tendencies (Furley & Memmert, 2021). Pertinent to the present research, a follow-up study using Active Appearance Modeling with Noldus FaceReader (Furley et al., 2021) indicated that facial expressions of professional darts players prior to poor performance showed slightly more arousal, sadness, and anxiety than before good

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performances, although no clear differences in individual muscle movement (i.e., action units) were evident using the automated FACS methodology. However, it seems feasible that the Active Appearance Modeling algorithm of Noldus FaceReader was not able to detect subtle muscle movements in the fairly neutral faces of professional darts players immediately prior to throwing darts (see Fig. 1). Hence, the present research attempted to scrutinize this initial research that utilized computerized facial action coding with the more time-consuming manual FACS. The FACS has become the standard tool for measuring the facial muscle movements building up facial expressions. More specifically, it is a reliable, anatomical system for describing all observable facial movements, breaking facial expressions down into individual small segments.

There has been a longstanding interest in human facial expressions (Piderit, 1858; Gratiolet, 1865) that is probably most prominently showcased by Charles Darwin's (1896) book "The Expression of the Emotions in Man and Animals" that proposed the argument that facial expressions are universal and innate characteristics of the human species. Pioneering psychology researcher William James assigned a pivotal role to facial expressions in the experience of emotion by arguing that facial expressions were not only indicative of emotional states, but actually caused or influenced experiencing an emotion via feedback processes from the facial musculature. This line of reasoning was elaborated upon by Sylvan Tomkins (1962, 1963) and has become prominent under the term facial feedback hypothesis. Although the role of facial expressions in emotions remains a much debated and highly controversial topic (e.g., Barrett, 2021; Coles et al., 2019; Cowen et al., 2021), people tend to generally agree that facial behavior can be regarded as an important source of information in a wide range of situations (Patterson et al., 2023).

FACS has hardly ever been used to measure facial muscle movement in televised sports competitions, which seems somewhat surprising given the abundance of video footage of sports and its emotional nature (Furley, 2019). In this respect, the present research has heuristic value by showcasing the utility of using FACS on televised video footage of athletes to investigate contingencies between cues in the face and performance tendencies. In addition, I attempt to compare manual FACS with published research using automated FACS coding with Noldus FaceReader (Furley et al., 2021). I consider this secondary goal to be important research in the field of NVB given the infancy of automated FACS and research highlighting differing results between automated and manual FACS (e.g., Skiendziel et al., 2019). While previous research has shown that under certain circumstances automated and manual FACS yield comparable results, in particular when using exaggerated, acted facial expressions of emotions (Lewinski et al., 2014; Skiendziel et al., 2019), this has rarely been tested in more subtle facial expressions in real world social interactions.

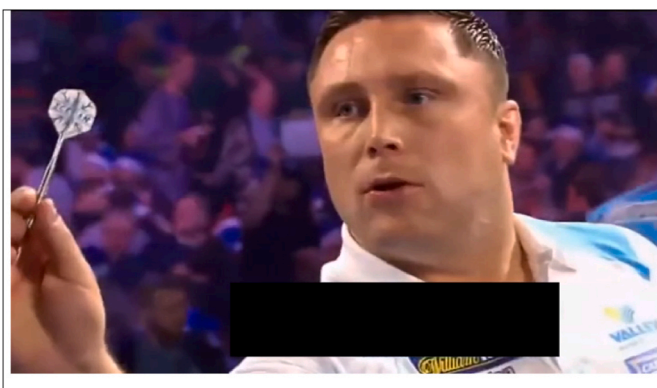


Fig. 1. Sample frame from the stimulus material used in the present research.

1. The present research

The present research builds on previous research in professional darts (Furley et al., 2021; Furley & Memmert, 2021) to test which facial muscle movements are associated with successful and unsuccessful performance using FACS as the Active Appearance Modeling algorithm of Noldus FaceReader hardly detected any facial muscle movement in the video data and therefore might have been affected by floor effects. To this end, the same stimulus set as in the automated coding study by Furley et al. (2021) was used that contrasted pre-performance facial expressions of professional darts players as a function of good and poor performance.

In developing FACS (Ekman and Friesen, 1976, 1978), two different types of judgments were distinguished, those measuring facial behavior and those making inferences about these behaviors. Ekman and Friesen have argued that measuring the activity of individual facial muscles (action units in FACS) can be considered a (mainly) descriptive analysis of behavior, whereas measuring emotional expressions like anger can be considered more of an inferential process (as in the EMFACS; a system derived from FACS to indicate emotions). The present research focused on the descriptive analysis of facial behavior and avoided making emotional inferences based on this behavior as the emotional connotation of facial muscle movements is a heavily debated topic with some scholars objecting to the theoretical tenet of humans possessing prototypical emotional facial expressions (i.e., universal anger or joy expressions; Barrett, 2021; Crivelli & Fridlund, 2018, 2019; Crivelli et al., 2016, 2017; Kollareth et al., 2023; Russell et al., 2003). Despite the debate about the existence of universal emotional facial expressions, scholars studying affective expressions tend to agree that the human face is an important source of information that can give observers accurate information about another person's internal states associated with arousal or tension (Patterson et al., 2023 for a recent overview). In this respect, facial cues have been described as an important part of the human capacity of mentalizing (Frith & Frith, 2003)—i.e., the capacity to understand ourselves and others in terms of intentional mental states, such as feelings, desires, wishes, attitudes, and goals.

Although most research using the FACS has involved posed facial expressions of emotions, it has been argued that the FACS can also be utilized as an immediate, objective, unobtrusive, and reliable measure of spontaneously generated expressions. First studies have shown the utility of FACS as a useful instrument to measure a wide range of spontaneous emotional (Sayette et al., 2001) and unemotional (Rojo et al., 2015) expressions. A few studies have therefore started to adopt FACS coding in the field of sports to examine (amongst other variables) the facial expressions of Judokas at the Olympic Games after they had won or lost a medal competition (Hwang & Matsumoto, 2014; Matsumoto et al., 2009; Matsumoto & Hwang, 2012; Matsumoto & Willingham, 2006, 2009; Tracy & Matsumoto, 2008). These studies provided evidence that athletes seem to show discrete facial muscle movements as an immediate response to victory or defeat, or stated differently, the FACS revealed unintended cues in the face that were associated with prior victory or defeat. These studies highlight the utility of using FACS coding in the context of professional sports (albeit these studies did not use televised footage, but high-resolution footage of athletes faces especially recorded for the purpose of the studies), which arguably seems to be an ideal context given the intense emotional nature of sports and the vast amount of spontaneous, real world facial expressions occurring in this context (Furley, 2019).

While there seems to be accumulating evidence that athletes' facial expressions after a performance differ depending on success and failure (but see Aviezer et al., 2012 for intense emotional displays), hardly any research has tested if facial expressions of athletes can also be used to predict performance. Regarding this research question, one study used FACS to code smiles of 152 Ultimate Fighting athletes one day before competition (Kraus & Chen, 2013). Results showed that increased smile intensity before an Ultimate Fighting match was associated with poorer

performance by the fighter showing the smile and better performance of the opponent. The present research attempted to extend this initial research by using FACS coding on televised sports footage to test the hypothesis if facial behavior immediately before throwing darts is associated with subsequent performance. Regarding this hypothesis, it is important to note that the informational value of any facial cue or set of cues is probabilistic at best (Hall et al., 2019) and I am not arguing for a causal relationship between pre-performance facial expressions and performance in sport. However, there is theoretical reason to believe that certain facial muscle behavior might be predictive of performance in sports. For example, meta-analyses indicated a connection between emotions and subsequent sport performance. Specifically, there is evidence that high levels of anxiety and physiological arousal can be negatively associated with athletic performance, particularly in situations requiring fine sensorimotor coordination (Craft et al., 2003; Jokela & Hanin, 1999). As affective states often show in a person's face, and previous research has shown that facial information of darts players was sufficient to infer accurate score tendencies (Furley & Memmert, 2021, Study 3), I hypothesized that facial muscle movement would differ prior to successful compared to unsuccessful performance.

In addition, and of particular relevance for the present research, early research has suggested a complicated relationship between muscular tension and performance in a variety of tasks (Courts, 1942; Freeman, 1938). This research suggests that depending on the performance muscle tension could be facilitative or debilitative to performance. Although, this line of research typically investigated muscle tension in musculature directly involved in subsequent performance, research in the area of motor control (e.g., Schmidt & Wrisberg, 2008 for an overview) suggests that more global muscle tone (i.e., muscle tension in the entire body) can impact on fine sensorimotor performance as in darts. Hence, it seems feasible that an increased muscle tone would show on darts players faces and this would be negatively correlated with darts performance.

Regarding the main objective of the present research, several lines of research point to the feasibility that facial muscle activity might be associated with differential performance in professional darts. More specifically, I hypothesized that higher tension in the face (i.e., higher action unit intensities) would be associated with lower darts scores. Regarding which AUs would differ and to what extent the study was explorative. Regarding the secondary goal, the present research compared action unit intensity between manual FACS coding and Noldus FaceReader to test if manual facial action coding was more sensitive to detect subtle differences in the fairly neutral televised faces of darts players compared to the FaceReader algorithm.

2. Method

2.1. Sample

The sample was the same as in Furley et al. (2021) and consisted of televised video recordings of 47 players of the first two rounds of the 2017 Professional Darts Corporation (PDC) World darts Championship. These included 29 matches of round one, whereby video footage of three matches of this round were missing, and all matches of round two (16). All matches were screened for the following two performance categories for each player. The high-performance category included all scores which contained at least two big triples (17–20 points) darts and at least a single (1–20 and 25) or better. Therefore, this category had a possible scoring range from 103 to 180 points. The low-performance category only included single scores or no scores and consisted of a scoring range from 0 to 75. Intentionally thrown bust-throws were excluded, even though these would have met the criterion scoring-wise. It was ensured that in both performance categories players were trying to maximize scores and not e.g. trying to set up a particular score or check out. A further requirement was to generate at least 10 throws in both performance categories over the course of these first two rounds. This

procedure made 50 out of 61 possible players eligible for the further course. Subsequently we numbered all eligible throws of these 50 players in the two performance categories and randomly selected ten high performance and ten low performance throws using the webpage www.random.org. The selected throws were then cut and edited with the DaVinci Resolve 16 software.

The created video clips showed the preparatory nonverbal behavior of one throw (Fig. 1), that is three consecutively thrown darts (see Furley & Memmert, 2021). The starting point of every video stimulus was determined when a player made a clear upwards movement with one dart in his throwing hand to initiate the beginning of the throwing motion after he had reached for a dart which he had held in his non-throwing hand or the arm was hanging loose on his side with the first dart in his hand. A throwing sequence ending point was determined as the last frame before the dart left the hand. Given that one frame is 33.33 ms long, this means that video information ended in the range of 0–33.33 ms before throwing the dart. All three pre-performance throwing sequences of a throw were merged together to one video clip. A video clip was not considered eligible for the study, if the footage of a selected throw did not show the player and his pre-performance nonverbal behavior of all three individual thrown darts within a throw. In that case the video clip was excluded and replaced by the subsequent number on the list of all eligible throws created by the webpage www.random.org. The broadcasted scoreboard and the dartboard, if it was shown in the split-screen, were blacked out (Fig. 1 provides as an example frame from a stimulus video) and the sound was muted.

As a result of this process the final data set consisted of 47 players with 10 valid video clips per performance category (940 video clips in total). All clips consisted of a resolution of 480 x 720 pixel with a frame rate of 30,00 fps (frames per second). The average duration of a video clip was 4.436s ($SD = 1.351$).

2.2. Procedure

Measures. For comparison purposes with previous research using the Active Appearance Modelling approach of FaceReader Version 7 (Van Kuilenburg et al., 2005), two FACS-certified coders manually coded the activation intensity of 20 distinct muscle groups in the face (i.e., 'action units'), which are used to generate facial expression (Ekman & Friesen, 1978). Hence, only these 20 action units were coded to compare the present analyses with the analysis of Furley et al. (2021) using Noldus FaceReader. Facial expressions in each recording were coded on a frame-by-frame basis. Coders of the video data were blind to performance condition.

Action unit (AU) intensities. Given the limited video quality and short duration of the televised darts players, it was decided to only score AU intensity and not onset and duration of AU (Cohn et al., 2007; Sayette et al., 2001; Scherer et al., 2019).

Two FACS-certified coders coded the intensity of the AUs (Table 1) on 6-point Likert scales (0 = none, 1 = trace, 2 = slight, 3 = marked/pronounced, 4 = severe/extreme, 5 = maximum).

Both coders were asked to provide one intensity value (i.e., mean of the frame-by-frame analysis) ranging from 0 to 5 for every one of the 940 videos and for every action unit, respectively. As there is no consensus on determining reliability of intensity in FACS coding ("FACS certification test, for instance, omits intensity differences altogether", Sayette et al., 2001, p. 170) the final Action Units Intensity score was the mean of both coders' individual intensity ratings for every darts player (47) in the high performance (10 videos) and low performance (10 videos) category (940 videos altogether). The rationale for this was to make the AU intensity variable more reliable given the fact that all video footage showed very limited facial muscle activity and coders were able to code individually without the need of interacting with one another to discuss disagreements, which would have been arduous given the 940 video clips every coder had to code. The difficulty of attaining reliability

Table 1
Specific Action Units and their Description.

Action Unit	Description
AU 1	Inner Brow Raiser (Muscular basis: frontalis, pars medialis)
AU 2	Outer Brow Raiser (Frontalis, pars lateralis)
AU 4	Brow Lowerer (depressor glabellae, depressor supercilii, and corrugator supercilii)
AU 5	Upper Lid Raiser (levator palpebrae superioris, and superior tarsal muscle)
AU 6	Cheek Raiser (Orbicularis oculi, pars orbitalis)
AU 7	Lid Tightener (Orbicularis oculi, pars palpebralis)
AU 9	Nose Wrinkler (Levator labii superioris alaeque nasi)
AU 10	Upper Lip Raiser (Levator labii superioris, caput infraorbitalis)
AU 14	Dimpler (Buccinator is the underlying muscle)
AU 15	Lip Corner Depressor (Depressor anguli oris)
AU 17	Chin Raiser (mentalis)
AU 18	Lip Pucker (incisivii labii superioris and incisivii labii inferioris)
AU 20	Lip Stretcher (risorius w/platysma)
AU 23	Lip Tightener (orbicularis oris)
AU 24	Lip Pressor (orbicularis oris)
AU 25	Lips Part (depressor labii inferioris, or relaxation of mentalis or orbicularis oris)
AU 26	Jaw drop (masseter; relaxed temporalis and internal pterygoid)
AU 27	Mouth Stretch (pterygoids and digastric)
AU 43	Eyes Closed (relaxation of Levator palpebrae superioris)

about the intensity of small facial motion in the televised footage has been discussed in the FACS literature (Sayette et al., 2001) and therefore the averaged intensity scores are the most appropriate measure for the objectives of the present research. Nevertheless, I checked the agreement between observers (i.e., inter-observer reliability) which showed high levels of agreement for the aggregate intensity scores (i.e., mean of the frame-by-frame analysis for the two performance categories of the 47 darts players) of all coded AUs, Correlations between coders demonstrated high consistency in the coding of all AUs intensity ($r_s = .76$ to $.96$, $p_s < .001$). In addition, the coders did not significantly differ when comparing their intensity scores in a series of Wilcoxon signed-rank tests on all 20 AUs (all $p > .5$).

Noldus FaceReader Action unit (AU) intensities. Face Reader Version 7 was used in the present research. Using Active Appearance Modelling (Van Kuilenburg et al., 2005), FaceReader detects the

activation intensity of 20 distinct muscle groups in the face (i.e., action units). FaceReader automatically computes continuous variables reflecting the intensity of the 20 AUs described in the previous section. I used the AUs intensities as published in Furley et al. (2021) for comparison purposes to the manual intensity codings. In contrast to the manual 6-point Likert scale (0 = none, 1 = trace, 2 = slight, 3 = marked/pronounced, 4 = severe/extreme, 5 = maximum) variable, AUs were exported as continuous variables with FaceReader ranging from 0 to 1. Values from .00 to .16 are classified as Trace, .16–.26 as Slight, .26–.58 as Pronounced, .58–.90 as Severe, and .90–1 as Max Intensity for every video frame. The Noldus AU intensity variable was computed as the mean of every analyzed frame in the respective video.

3. Results

The descriptive statistics are summarized in Table 2.

As would be expected based on anecdotal evidence from watching darts, the facial expressions of darts players showed extremely limited facial muscle movements in the exact moment immediately before throwing a dart. This clearly shows in the data, when considering the 6-point intensity rating scale with zero reflecting no muscle movement activity and 5 maximum activity (0 = none, 1 = trace, 2 = slight, 3 = marked/pronounced, 4 = severe/extreme, 5 = maximum). If there was any muscle movement activity at all, this was—generally speaking—no more than a trace. Nevertheless, a clear pattern of results emerged showing that if significant differences in certain action units emerged between good and poor performance throws, the difference was always characterized by less intense action unit activity. A 1x2 (good vs. poor performance) MANOVA including all 20 AUs using Wilks-Lambda revealed a significant main effect of performance category ($\Lambda = .262$; $F(20, 27) = 3.808$, $p < .001$, $\eta_p^2 = .738$), revealing less intense facial muscle movement activity across all action units. More specifically, there was significantly lower activity (see Table 2) in the inner brow raiser (AU1), brow lowerer (AU4), cheek raiser (AU6), dimpler (AU14), lip corner depressor (AU15), chin raiser (AU17), lip pressor (AU24), lips part (AU25), and blink/eyes closed (AU43). However, after Bonferroni adjusting the alpha level to $p = .0025$, only the inner brow raiser (AU1), the brow lowerer (AU4), the cheek raiser (AU6), lip corner depressor (AU15), the chin raiser (AU17), and lips part (AU25) remained

Table 2

Descriptive statistics as a function of High and Low Performance and Coding Procedure for all dart Throws per shown dart Player. Statistics were calculated on the mean values of the High and Low Performance of each Players dart Throws.

	FACS Coding					FaceReader (Furley et al., 2021)				
	High Performance		Low Performance		<i>p</i>	High Performance		Low Performance		<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
Action Unit Intensity (Micro)										
inner brow raiser (AU1)	0.193**	.403	0.327**	.439	.001	.0191	.0587	.0229	.0787	.566
outer brow raiser (AU2)	.448	.398	.433	.466	.630	.0045	.0276	.0034	.0220	.463
brow lowerer (AU4)	0.201**	.442	0.419**	.609	.001	.0504	.0911	.0428	.0871	.587
upper lid raiser (AU5)	.142	.137	.179	.195	.062	.0517	.1546	.0608	.1724	.465
cheek raiser (AU6)	0.402**	.462	0.820**	.697	.001	.0047	.0203	.0028	.0152	.326
lid tightener (AU7)	.088	.184	.077	.158	.714	.0500	.1092	.0523	.1068	.789
nose wrinkler (AU9)	.068	.171	.105	.209	.214	.0001	.0005	.0011	.0055	.241
upper lip raiser (AU10)	.116	.220	.112	.197	.669	.0143	.0676	.0283	.1085	.081
lip corner puller (AU12)	.324	.497	.323	.436	.626	.0260	.0656	.0376	.1156	.962
dimpler (AU14)	0.843*	1.038	1.082*	1.055	.004	.0417	.0834	.0397	.0668	.952
lip corner depressor (AU15)	1.147**	.266	1.256**	.303	.001	.2545	.2662	.2634	.2480	.135
chin raiser (AU17)	0.259**	.432	0.516**	.670	.001	.0715	.1116	.0687	.1033	.981
lip puckerer (AU18)	.108	.224	.121	.226	.782	.0023	.0088	.0041	.0143	.733
lip stretcher (AU20)	.063	.092	.085	.133	.581	.0025	.0062	.0016	.0052	.372
lip tightener (AU23)	1.553	.650	1.571	.549	.699	.0086	.0247	.0085	.0258	.543
lip pressor (AU24)	0.490*	.539	0.667*	.581	.005	.0324	.1090	.0417	.1179	.524
lips part (AU25)	0.154**	.194	0.402**	.387	.001	.0710	.1189	.0835	.1243	.164
jaw drop (AU26)	1.478	.671	1.454	.462	.657	0.0299*	0.0579	0.0412*	0.0698	.027*
mouth stretch (AU27)	1.346	.526	1.383	.447	.107	.0039	.0106	.0086	.0387	.925
blink/eyes closed (AU43)	1.183*	.268	1.255*	.324	.015	.1100	.1849	.1134	.1973	.485

* significant differences in Wilcoxon tests (** significant after Bonferroni correction).

significant.

A further 1x2 (good vs. poor performance) MANOVA including all 20 AUs scored by FaceReader using Wilks-Lambda did not reveal a significant main effect of performance category ($\Lambda = .506$; $F(20, 27) = 1.317$, $p = .249$, $\eta_p^2 = .494$). Although the MANOVA does not show a significant effect, Table 2 is indicative that Noldus values are also slightly lower (i. e., less intensity) in the high-performance category compared to the low performance category. The only significant difference (which was no longer significant after Bonferroni correction) showed less intense active in AU 26 (jaw drop). Given the different pattern of results between the Noldus FaceReader data and the manually coded FACS, Table 3 gives an overview of Spearman correlations coefficients between both analysis types for the means of the 47 darts players analyzed. As is evident from Table 3, there is very limited correlation between manually and automatically coded AUs. The lack of correlation suggests that only the manual FACS coding was sensitive enough to detect slight differences in facial muscle movements in the neutral faces of professional darts players depending on whether they were preceding good performances or bad performances.

4. Discussion

The aim of the present study was to use FACS on televised facial behavior of professional darts players immediately prior to throwing darts during the 2017 world darts championships to test if certain facial behavior was associated with successful and unsuccessful darts performance and to compare FACS to automated facial action coding using Noldus FaceReader. Specifically, it was hypothesized that higher tension in the face (i.e., higher action unit intensities) would be associated with lower darts scores. Data analysis over all action units revealed less intense facial muscle movement across the entire face, confirming this hypothesis. Regarding which AUs would differ and to what extent the study was explorative and specifically revealed that there was less intense action unit activity in the inner brow raiser (AU1), brow lowerer (AU4), cheek raiser (AU6), dimpler (AU14), lip corner depressor (AU15), chin raiser (AU17), lip pressor (AU24), lips part (AU25), and blink/eyes closed (AU43). After Bonferroni correction only the inner brow raiser (AU1), brow lowerer (AU4), the cheek raiser (AU6), lip corner depressor (AU15), the chin raiser (AU17), and lips part (AU25) remained significant. Hence, it seems that successful darts performance is associated with lower muscle activity around the eyes and the mouth.

Table 3

Spearman correlations between the manually coded FACS and the Noldus FaceReader data.

	N	Correlation FACS * Noldus			
		<i>rho</i>	<i>Lower CI</i>	<i>Upper CI</i>	<i>p</i>
Action Unit Intensity					
inner brow raiser (AU1)	47	.119	-.153	.411	.427
outer brow raiser (AU2)	47	-.102	-.385	.200	.492
brow lowerer (AU4)	47	-.078	-.365	.222	.601
upper lid raiser (AU5)	47	.063	-.236	.352	.673
cheek raiser (AU6)	47	.028	-.269	.321	.851
lid tightener (AU7)	47	.029	-.269	.321	.848
nose wrinkler (AU9)	47	.160	-.142	.435	.283
upper lip raiser (AU10)	47	-.054	-.344	.245	.718
lip corner puller (AU12)	47	.168	-.134	.441	.258
dimpler (AU14)	47	.447	.175	.656	.002
lip corner depressor (AU15)	47	-.111	-.393	.191	.458
chin raiser (AU17)	47	.155	-.147	.431	.298
lip puckerer (AU18)	47	.157	-.145	.432	.292
lip stretcher (AU20)	47	-.060	-.349	.240	.690
lip tightener (AU23)	47	.213	-.088	.478	.152
lip pressor (AU24)	47	-.003	-.298	.292	.984
lips part (AU25)	47	.217	-.084	.481	.143
jaw drop (AU26)	47	.391	.109	.616	.007
mouth stretch (AU27)	47	-.093	-.378	.208	.535
blink/eyes closed (AU43)	47	.140	-.162	.418	.349

Regarding the secondary goal, the present research compared action unit intensity between manual FACS coding and Noldus FaceReader indicating that the manual descriptive analysis of AUs using FACS was more sensitive in detecting subtle differences in the mainly neutral faces of Darts players compared to automated coding of the same AUs. The lack of correlation between manual and automated coding in the context of this study suggests that manual coding was more sensitive in detecting subtle differences in fairly neutral televised faces as FaceReader detected even less activity in darts players faces compared to FACS coders.

Although, the FACS was developed alongside the theoretical perspective of Basic Emotion Theory (Ekman, 1992; Keltner et al., 2019), which argues for universal basic emotions that are associated with distinct facial expressions, the present pattern of results does not suggest that a distinct facial expression exists in professional darts players that is predictive of good or poor darts performance. Hence, when observers of darts competitions were shown to be able to accurately infer performance tendencies based on thin-slices of pre-performance nonverbal behavior (Furley & Memmert, 2021), this was likely not informed by distinct facial expressions coinciding with good or poor performance. More likely, observers picked up a general pattern of less intense facial muscle activity that seems to be a valid cue in estimating subsequent performance. This pattern might further reflect previous literature suggesting that low levels of muscular tension and/or arousal and anxiety prior to fine sensorimotor performance in sports are associated with better performance than high levels of tension and/or arousal and anxiety (Furley et al., 2023; Schmidt & Wrisberg, 2008).

Although Furley et al. (2021) and the present analysis did not find corresponding differences in individual AU intensities using Noldus FaceReader, the automated coding did suggest slightly more arousal, sadness, and anxiety before poor performance as compared to good performances. In this respect the findings from the present research might be regarded as comparable to the automated coding study as reflected in the significant MANOVA indicating less intense facial muscle movement activity before successful performances compared to less successful performances. Regarding the individual AU differences found in the present study as a function of performance, the results did not resemble the findings of the automated coding procedure reported in Furley et al. (2021) as there were no corresponding differences in the 20 manually coded AUs and the computer coded AU. The reason for this might arguably be that the video recordings showed very little AU activity in general and the traces of activity in parts of the analyzed videos might have often gone undetected by the FaceReader algorithm. Further, previous related research has indicated differing results between automated and manual FACS (e.g., Skiendziel et al., 2019), although comparable results between automated and manual FACS have been obtained when using exaggerated, acted facial expressions of emotions (Lewinski et al., 2014; Skiendziel et al., 2019). Initial sports related research using Noldus FaceReader (Hetland et al., 2018, 2019) has shown the utility of FaceReader in tapping into emotions of athletes during competition, although emotional expressions were shown to be largely unrelated to self-report measures of emotions. Hence, I do not want to question the merit of Noldus FaceReader Version 7 per se, but the present research shows that manual FACS coding seems to be able to capture finer differences in neutral televised faces of darts players. Further research is needed regarding the utility of both manual and automated facial coding procedures in reliable capturing subtle real-world facial expressions.

Nevertheless, some interesting results emerged from the present research that might serve as a starting point for future research on facial behavior in real world social interactions. For example, Figs. 2 and 3 illustrate specific facial behavior that was (on average) more pronounced before less successful performances in Darts.

The individual AUs illustrated in Figs. 2 and 3 should be scrutinized in further research in the area of sport or other performance domains as they might constitute an overt visual cue that is indicative of a suboptimal internal state that is associated with poor sensorimotor



Fig. 2. Sample frames from the stimulus material used in the present research to highlight significant differences in AU in the mouth/chin/cheek area.



Fig. 3. Sample frames from the stimulus material used in the present research to highlight significant differences in AU in the eyes area.

performance. It seems feasible that the facial expressions depicted in Figs. 2 and 3 might signal high levels of physiological arousal, tension or anxiety that have been shown to be associated with less successful performance in sport (Furley et al., 2023 for a review). Although, the present research used a descriptive instead of inferential analysis of facial muscle movements and deliberately avoided making inferences about underlying emotional states, the depicted facial expressions in Figs. 2 and 3 do show some overlap with facial expressions of heightened arousal, tension, and anxiety (Ekman & Friesen, 1976, 1978).

From an applied perspective, the results might suggest that suboptimal levels of arousal, tension or anxiety show on athletes' faces. Therefore, athletes or coaches might want to pay attention to similar cues as depicted in Figs. 2 and 3 and if detected might want to attempt to reduce facial tension in these AU. Although the facial feedback hypothesis (Coles et al., 2019 for an overview) has been a very controversial topic in the recent psychological literature, it seems to be a viable hypothesis that modifying facial muscle activity in the entire face or in individual AUs (e.g., the ones in Figs. 2 and 3) might affect behavior and/or performance in a variety of contexts. In this regard, I do consider the findings from the present research rather preliminary, but think that this research offers fruitful potential to use FACS coding or other bodily coding techniques (e.g., Furley & Roth, 2021) to investigate contingencies between facial and bodily cues and other variables of interest like performance in sports.

The present research has some notable strengths and weaknesses. The major strength of the research is that it is one of the few studies that has attempted to utilize the arduous FACS coding on a carefully chosen sample of televised sports videos. Whereas, first research has applied similar methodology on intense emotional expressions at the concluding moment of Olympic Judo competitions (Hwang & Matsumoto, 2014; Matsumoto et al., 2009; Matsumoto & Hwang, 2012; Matsumoto & Willingham, 2006, 2009; Tracy & Matsumoto, 2008), the present research was the first to suggest that FACS might have further utility in detecting subtle differences in facial behavior in televised sport broadcasts. Moreover, I consider it a strength that results were based on

randomly chosen video clips that showed 10 respective good performances and 10 poor performances of the same athlete. This within-subject approach represents a powerful methodology to detect subtle facial differences as a function of performance. Finally, the present research is one of the first studies to compare automated and manual facial coding procedures with real life video clips instead of artificially created stimulus material. However, a limitation of the present methodology was that the video broadcasted material was of limited quality and only had 30 frames/second, which is problematic for FACS coding (Ekman & Friesen, 1978). In addition, videos were very short and showed very little muscle activity in general, which makes floor effects more likely. Also, the camera perspective, length of videos was not standardized as we had to work with the televised video material. Therefore, it is not clear if other information in the video material influenced the coding procedure. Moreover, it is unclear to what extent darts players might deliberately be trying to modify or mask their facial expressions in an attempt of impression management (e.g., Furley, 2023) and thereby try to hide felt anxiety and tension. However, this is likely the case in the high- and low-performance category and results show that felt internal states "leak to the outside" via facial muscle movements of professional darts players. Finally, it is not clear if the present findings are limited to professional darts or might transfer to other performance situations.

In conclusion, the present research highlights that the established methodology of FACS might have utility on televised sports videos. Given the emotional nature of this context and the amount of video documentation, this methodological approach arguably offers great potential in adding to the literature of nonverbal behavior and emotions and might help to resolve some of the big controversies in this field (Barrett, 2021; Cowen et al., 2021; Keltner et al., 2019; Lindquist et al., 2022; Patterson et al., 2023). Besides this methodological contribution, the present research demonstrated that less intense facial muscle activity in the moments before throwing darts in a professional competition is associated with better performance, in particularly in certain AU around the mouth and the eyes of the athletes.

Research disclosure statements

The authors declare that 1) (a) the total number of excluded observations and (b) the reasons for making these exclusions have been reported in this manuscript; 2) that all independent variables or manipulations, whether successful or failed, have been reported in the manuscript; 3) that all dependent variables or measures that were analyzed for this article's target research question have been reported in the manuscript/online supplements; 4) that (a) how sample size was determined and (b) our data-collection stopping rule have been reported in this manuscript.

Participant consent statement

Not applicable.

Permission to reproduce material from other sources

Not applicable.

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All data will be made available at reasonable request to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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